Partial Derivatives

Function of multiple variable

Function of two variable: A function of two variable, $x$and y, is a rule that assigns a unique real number $f(x, y)$ to each point (x, y) in some set D in the xy −plane.

Function of three variable: A function of two variable, x, y and z, is a rule that assigns a unique real number $f(x, y, z)$ to each point (x, y, z) in some set D in three-dimensional plane.

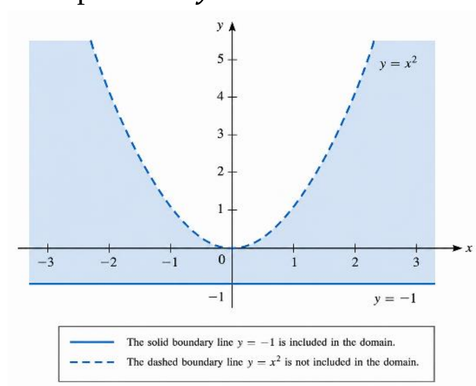
Example-1:

Let $f(x, y) = \sqrt{y+1} + \ln(x^2 - y)$. Find $f(e, 0)$ and sketch the natural domain of the function.

Solution: By substitution,

$$f(e, 0) = \sqrt{1} + \ln(e^2) = 1 + 2 = 3$$

To find the natural domain of f , $\sqrt{y+1}$ we note that is defined only when $y \geq -1$, while $\ln(x^2 - y)$ is defined only when $0 < (x^2 - y)$ or $y < x^2$. Thus, the natural domain of f consists of all points in the xy-plane for which $-1 \leq y < x^2$. To sketch the natural domain, we first sketch the parabola $y = x^2$ as a "dashed" curve and the line $y = -1$ as a solid curve. The natural domain of f is then the region lying above or on the line $y = -1$ and below the parabola $y = x^2$.



Example-2: Let $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$. Find $f\left(0, \frac{1}{2}, -\frac{1}{2}\right)$ and the natural domain of f.

Solution: By substitution,

$$f\left(0, \frac{1}{2}, -\frac{1}{2}\right) = \sqrt{\frac{1}{2}}$$

Because of square root sign we must have, $0 \leq 1 - x^2 - y^2 - z^2$ in order to have real value of $f(x, y, z)$. Rewriting the form

$$x^2 + y^2 + z^2 \leq 1$$

Therefor we see that the natural domain of f consists of all points on or within the sphere

$$x^2 + y^2 + z^2 = 1$$

Example-3: In each part describe the graph of the function in xyz-cordinate system.

a) $f(x,y) = 1 - x - \frac{1}{2}y$
b) $f(x,y) = \sqrt{1 - x^2 - y^2}$.

Solution: (a) By the definition, the graph of given function is the graph of the equation

$$z = 1 - x - \frac{1}{2}y$$

Which is a plane. A triangular portion of this plane can be sketched by plotting the intersections with the coordinate axes and joining them with the line segments. The intersections with the coordinates are $(1,0,0)$, $(0,2,0)$, $(0,0,1)$ respectively.

(b) By the definition, the graph of given function is the graph of the equation

$$z = \sqrt{1 - x^2 - y^2} \qquad (1)$$

After squaring both sides we get

$x^2 + y^2 + z^2 = 1$

Which represents a sphere of radius 1, centered at origin. Since (2) imposes the condition $z \geq 0$, the graph is just the upper hemisphere.

Limit and Continuity:

Limit of functions of two variables: Let *f* be a function of two variables, and assume that *f* is defined at all points of some open disk centered at (x_0, y_0), except possibly at (x_0, y_0). We will write

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$$

if given any number $\varepsilon > 0$, we can find a number $\delta > 0$ such that $f(x,y)$ satisfies

$$|f(x,y) - L| < \varepsilon$$

whenever the distance between (x,y) and (x_0, y_0) satisfies

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

Limit of functions of three variables: Let *f* be a function of three variables, and assume that *f* is defined at all points of some open disk centered at (x_0, y_0, z_0), except possibly at (x_0, y_0, z_0). We will write

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z) = L$$

if given any number $\varepsilon > 0$, we can find a number $\delta > 0$ such that $f(x,y)$ satisfies

$$|f(x,y,z) - L| < \varepsilon$$

whenever the distance between (x,y) and (x_0, y_0) satisfies

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} < \delta$$

Example-4: Evaluate $\lim_{(x,y)\to(-1,2)} \frac{xy}{x^2+y^2}$

Solution: $\lim_{(x,y)\to(-1,2)} \frac{xy}{x^2+y^2}$

$$= \frac{(-1)2}{1+4}$$

$$= \frac{-2}{5}$$

Example-5: Evaluate $\lim_{(x,y)\to(1,4)} (5x^3y^2 - 9)$

Solution: $\lim_{(x,y)\to(1,4)} (5x^3y^2 - 9)$

$$= \lim_{(x,y)\to(1,4)} (5x^3y^2) - \lim_{(x,y)\to(1,4)} 9$$

$$= 5[\lim_{(x,y)\to(1,4)} x^3 \lim_{(x,y)\to(1,4)} y^2] - 9$$

$$= 5 \times 1 \times 16 - 9 = 71$$

Theorem:

(a) If $f(x,y) \to L$ as $(\mathrm{x},\mathrm{y}) \to (x_0, y_0)$, then $f(x,y) \to L$ as $(\mathrm{x},\mathrm{y}) \to (x_0, y_0)$, along any smooth curve.

(b) If the limit of $f(x,y)$ fails to exist as $(\mathrm{x},\mathrm{y}) \to (x_0, y_0)$ along some smooth curve, or if $f(x,y)$ has different limits as $(\mathrm{x},\mathrm{y}) \to (x_0, y_0)$ along two different smooth curves, then the limit of $f(x,y)$ does not exist as $(\mathrm{x},\mathrm{y}) \to (x_0, y_0)$.

Example-6: $\lim_{(x,y)\to(0,0)} -\frac{xy}{x^2+y^2}$. Does the limit exist?

Solution: $\lim_{\substack{(x,y)\to(0,0)\\ along\ x=0}} -\frac{xy}{x^2+y^2} = 0$

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ y=0}} -\frac{xy}{x^2+y^2} = 0$$

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ y=x}} -\frac{xy}{x^2+y^2} = -\frac{1}{2}$$

We found two different smooth curves along which this limit had different values. Therefor the limit doesn't exist.

Example-9: Evaluate $\lim_{(x,y)\to(0,0)} \frac{x-y}{x^2+y^2}$

Solution:

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ x=0}} \frac{x-y}{x^2+y^2} = \lim_{y\to 0} \frac{-y}{y^2} = -\infty$$

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ y=0}} \frac{x-y}{x^2+y^2} = \lim_{y\to 0} \frac{x}{x^2} = \infty$$

Therefor $\lim_{(x,y)\to(0,0)} \frac{x-y}{x^2+y^2}$ doesn't exist.

Example-10: Evaluate $\lim_{(x,y)\to(0,0)} \frac{x+y}{2x^2+y^2}$

Continuity of functions of two variable: A function $f(x,y)$ is said to be ***continuous at*** (x_0,y_0) if $f(x_0,y_0)$ is defined and if

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0,y_0)$$

In addition, if f is continuous at every point in an open set D, then we say that f is continuous on D, and if f is continuous at every point in the xy-plane, then we say that f is continuous everywhere.

Example-7:

$$f(x,y) = \begin{cases} \dfrac{\sin(x^2+y^2)}{x^2+y^2} & if\ \ (\mathrm{x,y}) \neq (0{,}0) \\ 1 & if\ (\mathrm{x,y}) = (0{,}0) \end{cases}$$

Show that f is continuous at $(0,0)$

Solution: To show that

$$f(x,y) = \begin{cases} \dfrac{\sin(x^2+y^2)}{x^2+y^2} & if\ \ (\mathrm{x,y}) \neq (0{,}0) \\ 1 & if\ (\mathrm{x,y}) = (0{,}0) \end{cases}$$

Is continuous at $(0,0)$ we must to proof that $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 1$

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ x=0}} \frac{\sin(x^2+y^2)}{x^2+y^2} = \lim_{y\to 0} \frac{\sin y^2}{y^2} = 1$$

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ y=0}} \frac{\sin(x^2+y^2)}{x^2+y^2} = \lim_{x\to 0} \frac{\sin x^2}{x^2} = 1$$

$$\lim_{\substack{(x,y)\to(0,0)\\ along\ y=x}} \frac{\sin(x^2+y^2)}{x^2+y^2} = \lim_{x\to 0} \frac{\sin(2\,x^2)}{2x^2} = 1$$

Therefor $\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} = 1$

Therefor $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 1$

Hence the function continuous at $(0,0)$.

Example-10:

$$f(x,y) = \begin{cases} \dfrac{x+y}{2x^2+y^2} & if\ \ (\mathrm{x,y}) \neq (0,0) \\ 0 & if\ (\mathrm{x,y}) = (0,0) \end{cases}$$

Show that f is discontinuous at $(0,0)$

Solution: To show that

$$f(x,y) = \begin{cases} \dfrac{x+y}{2x^2+y^2} & if\ \ (\mathrm{x,y}) \neq (0,0) \\ 0 & if\ (\mathrm{x,y}) = (0,0) \end{cases}$$

Is continuous at $(0,0)$ we must to proof that $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 1$

$$\lim_{\substack{(x,y)\to(0,0) \\ along\ x=0}} \frac{x+y}{2x^2+y^2} = \lim_{y\to 0} \frac{y}{y^2} = \infty$$

$$\lim_{\substack{(x,y)\to(0,0) \\ along\ y=0}} \frac{x+y}{2x^2+y^2} = \lim_{y\to 0} \frac{x}{2x^2} = \infty$$

$$\lim_{\substack{(x,y)\to(0,0) \\ along\ y=-x}} \frac{x-x}{2x^2+x^2} = 0$$

Therefor $\lim_{(x,y)\to(0,0)} \frac{x+y}{2x^2+y^2}$ doesn't exist

Therefor f is discontinuous at $(0,0)$.

Chain Rule:

Chain Rule for Derivative: If $x = x(t)$ and $y = y(t)$ are differentiable at t, and if $z = f(x,y)$ is differentiable at the point $(x,y) = (x(t), y(t))$, then $z = (x(t), y(t))$, is differentiable at t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x,y)

Example-11: Suppose that, $z = x^2 y$, $x = t^2$, $y = t^3$. Find $\frac{dz}{dt}$ by using chain rule.

Solution:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$\frac{dz}{dt} = 2xy\frac{dx}{dt} + x^2\frac{dy}{dt}$$

$$\frac{dz}{dt} = 2xy\frac{d}{dt}(t^2) + x^2\frac{d}{dt}(t^3)$$

$$\frac{dz}{dt} = 4txy + 3t^2x^2$$
$$\frac{dz}{dt} = 4t^6 + 3t^6 = 7t^6$$

Example-12: Suppose that,

$$w = \sqrt{x^2 + y^2 + z^2}, x = cos\Theta, y = sin\Theta, z = tan\Theta$$

Use the chain rule to find $\frac{dw}{d\Theta}$.

Solution:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{d\Theta} + \frac{\partial w}{\partial y}\frac{dy}{d\Theta} + \frac{\partial w}{\partial z}\frac{dz}{d\Theta}$$

$$\frac{dw}{dt} = \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2x)(-sin\Theta) + \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2y)(cos\Theta)$$
$$+ \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2}(2z)(sec^2\Theta)$$

Chain Rule for Partial Derivative: If $x = x(u, v)$ and $y = y(u, v)$ have first-order partial derivatives at the point (u, v), and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(u, v), y(u, v))$, then $z = (x(u, v), y(u, v))$ has first order partial derivatives at the point (u, v)given by

$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$ and $\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$

Example-13: $z = e^{xy}, x = 2u + v, y = {u}/{v}$

Find $\frac{\partial z}{\partial u}$ *and* $\frac{\partial z}{\partial v}$ by using chain rule.

Solution: $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$

$$= \frac{\partial}{\partial x}(e^{xy})\frac{\partial}{\partial u}(2u + v,) + \frac{\partial}{\partial y}(e^{xy})\frac{\partial}{\partial u}({u}/{v})$$

$$= 2ye^{xy} + xe^{xy}\frac{1}{v}$$
$$= (2y + \frac{x}{v})e^{xy}$$
$$= ({2u}/{v} + \frac{2u + v}{v})e^{({u}/{v})(2u+v)}$$
$$= (\frac{4u}{v} + 1)e^{({u}/{v})(2u+v)}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$
$$= \frac{\partial}{\partial x}(e^{xy})\frac{\partial}{\partial v}(2u + v,) + \frac{\partial}{\partial y}(e^{xy})\frac{\partial}{\partial v}({u}/{v})$$
$$= ye^{xy} + xe^{xy}(-\frac{u}{v^2})$$
$$= (y - \frac{xu}{v^2})e^{xy}$$
$$= [\frac{u}{v} - (2u + v)\frac{u}{v^2}]e^{({u}/{v})(2u+v)}$$

$= -\frac{u^2}{v^2} e^{(u/v)(2u+v)}$

Example-14: $w = e^{xyz}, x = 3u + v, y = u^2 v$

Find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ by using chain rule

Solution: $\frac{\partial w}{\partial u} = \frac{\partial}{\partial x}(e^{xyz})\frac{\partial}{\partial u}(3u + v) + \frac{\partial}{\partial y}(e^{xyz})\frac{\partial}{\partial u}(u^2 v)$